FAST- National University of Computer & Emerging Sciences, Karachi.
School of Computing,
Mid Term II Examination, Spring 2023
3rd April 2023, 10:00 am – 11:00 am

Course Code: CS-2001	**Course Name:** Data Structures
Instructors: Anam Qureshi, Safia Baloch, Nida Munawwar, and Mafaza Mohi	
Student Roll No:	**Section:**

Instructions:
- Except your Roll No and Section, DO NOT WRITE anything on this paper.
- Return the question paper with your answer sheet.
- Read each question completely before answering it. There are 3 **questions on 2 pages.**
- In case of any ambiguity, you may make assumptions but your assumption must not contradict any statement in the question paper.
- All the answers must be solved according to the SEQUENCE given in the question paper, otherwise points will be deducted.

Time Allowed: 60 minutes **Maximum Points**: 15

Advanced Sorting Techniques	
Question No. 1 [CLO 1]	**[Time: 20 minutes] [Points: 5]**

A school is trying to take an annual photo of all the students. The students are asked to stand in a single line in non-decreasing order by height. Let this ordering be represented by the integer array "expected" where expected[i] is the expected height of the ith student in line.
You are given an integer array "height" representing the current order in which the students are standing in. Each height[i] is the height of the ith student in line.
Write a function that returns the number of indices where heights[i]!=expected[i]

For example
Input: heights = [1,1,4,2,1,3]
Output: 3 indices do not match.
Explanation:
heights: [1,1,4,2,1,3]
expected: [1,1,1,2,3,4]
Indices 2, 4, and 5 do not match.

Stack and Queue	
Question No. 2 [CLO 3]	**[Time: 20 minutes] [Points: 5]**

You are working for an e-commerce company that sells books online. The company has a database of book titles in which some book titles have duplicate letters. As a software developer, you have been tasked with creating a program that takes a book title as input and removes any duplicate letters in the title so that each letter appears only once. Your task is to write a program that takes a book title as input and uses a stack to remove any duplicate letters from the title. The program should output the resulting title with no duplicate letters.

For example, if the input book title is "maulajutt", the program should output "ajltmu". If the input book title is "thehungergames", the program should output "aeghmnrstu".
Note that the program assumes that the input string contains only lowercase letters.

Binary Tree	
Question No. 3 [CLO 2]	**[Time: 20 minutes] [Points:5]**

Suppose you are given an array of integers representing the level-order traversal of a complete binary tree, where the left child of a node is at 2*i + 1. The right child of a node is at 2*i + 2, where i is the index of the current node in the array. Write a function to calculate the sum of all the leaf elements in the binary tree.

For example, given the array [0, 1, -1, 9, 6, 4,100, 56, 23], the corresponding complete binary tree is shown below:

```
       0
     /    \
    1      -1
   / \     / \
  9  6    4   100
 /    \
56    23

     sum= 183
```

******************** Good Luck *******************